

Nicolò Migliore

Biomechanics Engineer

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EDUCATION

Polytechnic of Turin

Master's degree in Biomedical Engineering – biomechanics course

Turin, Italy

Sep 2023 - Jul 2026

- Master Thesis | **HEAL ITALIA Foundation project** in collaboration with UniMORE

Oct 2025 - Jul 2026

Title: “**Advancing In Silico Modeling for Hemodialysis: From Patient-Specific Prediction of Uremic Toxin Kinetics to Hemodialyzer Performance Analysis**”

Development and optimization of a mechanistic mathematical model to predict protein-bound uremic toxin (PBUT) kinetics during on-line hemodiafiltration, leveraging clinical data to support patient-specific treatment monitoring. Complementing this work, a comparative analysis of two commercial high-flux hemodialyzers was carried out using computational fluid dynamics (CFD), specifically investigating the blood compartment to assess the influence of device design on blood flow dynamics. To ensure a rigorous study, statistical analyses were conducted to process and evaluate the acquired data.

Relevant courses: Innovations in cardiovascular and respiratory engineering | Mechanics applied to biomedical systems | Numerical modeling techniques | Biomechanics of the cardiovascular system | Biomedical instrumentation and safety | Design of prostheses | AI in medicine | Biomedical signal processing

University of Palermo

Bachelor's degree in Biomedical Engineering

Palermo, Italy

Sep 2020 - Jul 2023

- Thesis: Wireless powered ventricular assist device (VAD)

PROJECTS

The impact of Endovascular Aneurysm Repair (EVAR) on abdominal aortic fluid dynamics using CFD. By comparing pre- and post-operative patient-specific CAD models under various flow conditions, the study demonstrated EVAR's key role in restoring physiological flow and reducing risks of aneurysm progression or rupture.

Biomechanical Structural Analysis using ANSYS & LS-DYNA: Conducted advanced finite element analysis (FEA) to evaluate the structural behavior of jaw and tooth. Performed stress and deformation analyses on 3D mandibular models under realistic loading conditions, along with modal and transient analyses to assess frequency response. Executed dynamic chewing simulations in LS-DYNA to evaluate interface stress distribution within reconstructed tooth assemblies.

Multidomain Bio-Modeling & Gait Kinematics: the project evaluates athletic jump biodynamics, equivalent RLC electrical modeling of respiratory dynamics under physiological and pathological conditions, and forearm muscle mechanics. Developed pipeline to process time-series data from wearable IMU sensors (APDM Opal), successfully quantifying spatio-temporal gait parameters across multiple walking speeds to assess biomechanical performance and gait variability.

Similarity analysis and pharmacophore modeling of bioactive food compounds for therapeutic applications: an insight on Breast Cancer

Calcification process in biological heart valves (BHV): polyphenol-based treatments to mitigate calcification

AI-Driven Prediction of Mechanical Ventilation Weaning: developed a ML pipeline to predict extubation outcomes based on patient clinical data. Implemented Genetic Algorithms for feature selection and utilized hierarchical clustering and Self-Organizing Maps (SOMs) for balanced dataset partitioning. Trained, optimized, and cross-compared multiple classifiers: kNN, Naive Bayes, and MLP with GA-optimized architectures, to maximize predictive balanced accuracy.

TECHNICAL SKILLS

Languages: Italian (native), English: IELTS Academic score 6.0 (equivalent to CEFR B2), Spanish (beginner)

Programming & Data processing: Python, Matlab, GitHub, Simulink, NI Labview

FEA & CFD: ANSYS LS-Dyna, Fluent, Paraview, Mimics, Patran, Nastran

CAD & 3D Modeling: Solidworks, AutoCAD, Rhinoceros, PTC Creo

Others: MS Office, SIMvascular, PIVlab, MOE, PSpice, Regulatory knowledge EU MDR 2017/745

ADDITIONAL INFORMATION

Exchange Program at Archbishop McCarthy High School, Miami, USA (Feb 2018).

“**Guida al Project Management**” certification issued by the Polytechnic of Turin | My Open edX (Sep 2025).

Acquired hands-on proficiency in experimental testing and data acquisition (DAQ) by configuring physical sensor networks with load cells and LVDTs.

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